

Proposal

Instructions:

The proposal accounts for 10% of the final project grade. Proposal should be limited to 2 pages. Submit your proposal by the due date indicated on the course syllabus. To begin with, you can study an application problem for which parallel computing has been used. It's best if you can pick a problem from your own research area. Describe the application, the use of parallelism, and your expectation of its success, strengths/weaknesses, and challenges. Some specific details to consider include the following:

Project Proposal Requirements

You must submit a brief project proposal outlining the following:

1. Scientific Relevance

- Clearly describe the problem you're addressing.
- Why is this problem scientifically important or computationally demanding?

2. Benchmarking

- Are existing benchmark results available?
- If so, how do these compare with your simulations or expected outcomes?

3. Parallel Platform

- What parallel computing model will you use?
(e.g., shared memory, distributed memory, GPU acceleration)

4. Performance Analysis

- Have you examined how well the application performs relative to the peak performance of the system or cluster you'll be using?

5. Scalability

- How does your application's performance scale with the number of processors or computing nodes?

6. Bottlenecks

- Can you identify any performance bottlenecks in the application?
- How might they be addressed or mitigated?

Not all of these details will be available for all applications. But for your final project, you must describe what you find worthwhile regarding the implementation.

Suggested Project Topics

You are welcome to propose your own topic, but below is a list of example projects:

- Parallelizing a Particle Simulation (Particles move in space according external, nearby, and far field forces)
- Finite Element/Finite Difference Domain Decomposition (Parallelization of a Finite Element/Finite Difference Solver for the Heat Equation)
- Parallelized Singular Value Decomposition
- Monte Carlo Simulation
- Pseudospectral Methods (using FFTW)
- Adaptive Poisson Solver and Graph Partitioning for Load Balancing
- Parallelized Stochastic Gradient Descent
- Fast Parallel Iterative Matrix Diagonalization
- Parallel Preconditioned Conjugate Gradient
- Parallel Solvers for Tridiagonal Linear Systems

Projects will be evaluated based on technical rigor, the extent of parallelization, performance analysis, and clarity of communication.

Project and Presentation

The final project focuses on analyzing and understanding parallel performance. The core objective is to explore, measure, and assess how parallelism contributes to computational efficiency. Your project must involve significant parallel computing effort—this can include implementing a method from a research paper, adapting an existing codebase, or applying parallel techniques to a computationally demanding application.

You will submit both a written report and a presentation, which together account for 25% of your final grade. The final report must be limited to 10 pages, single-column format, using standard LaTeX conventions. The project presentation is due by the final exam date.

Final Report Requirements

Your report must include the following components:

1. Hypothesis and Objectives

- Clearly state the main question your project seeks to explore or answer.
- What is your hypothesis regarding the performance or scalability of the application?
- What might be the results of this investigation, and why are they significant?

2. Context

- Will you be working independently or as part of a team?
- Is this project part of a larger research initiative, a collaboration, or an extension of prior coursework?
- Will the methods or results be applicable in other academic contexts or classes?

3. Related Work

- Identify 3 or more prior works that motivate, inform, or contrast with your project.

- These could include research papers, existing tools, or previous implementations in the field.

4. Empirical Methodology

- Describe what experiments or benchmarks you will run.
- Detail your use of parallel platforms (e.g., shared memory, distributed systems, GPUs).
- What resources (hardware, software, datasets) are needed for your investigation?
- How will you measure performance, and what metrics will be used (e.g., speedup, efficiency, scalability)?

5. Performance Analysis

- Discuss how your application performs in practice relative to the theoretical or peak performance of the system.
- Address scalability: How does performance change with increasing computational resources?
- Identify and analyze bottlenecks. Are there known or suspected issues that limit parallel performance? How might they be mitigated?

6. Challenges and Limitations

- Describe any anticipated obstacles (e.g., technical limitations, time constraints, system access).
- How might these affect the depth or scope of your study?

Citation and Academic Integrity

You must properly cite all sources of external ideas, data, software, algorithms, or text. This includes:

- Research papers that motivate your project or provide algorithmic foundations.
- Existing software libraries or codebases that you adapt, benchmark, or analyze.

- Benchmarks, datasets, or computational tools used in your experiments.

Use a consistent citation format (e.g., IEEE, ACM, APA) and include a references section at the end of your report. If you adapt code, note this clearly in your comments and/or documentation. Failure to properly attribute work may result in an academic integrity violation.

Presentation Content Criteria

Your presentation is an important part of your final submission and will be evaluated for clarity, communication, and technical depth. The presentation should be approximately 10 in length and should include the following elements:

- **Clear Statement of Purpose:** Clearly introduce your research question or hypothesis.
- **Background and Context:** Briefly summarize relevant prior work and the motivation for your project.
- **Technical Description:** Explain your approach, including the parallel computing model used.
- **Performance Results:** Present key findings, including graphs or tables showing performance, scalability, and bottlenecks.
- **Challenges and Lessons Learned:** Reflect on any difficulties and insights gained during the project.
- **Conclusion:** Summarize the main takeaways and suggest directions for future work.

Rubric

	Exemplary	Satisfactory	Needs Improvement	Unsatisfactory
1. Practical Content & Implementation	Demonstrates creativity and depth; implementation is technically sound, well-tuned, and clearly organized. Significant effort is evident.	Reasonably creative and complete; implementation functions as intended with moderate optimization and clarity.	Basic implementation with limited optimization or insight; may lack modularity or clarity.	Incomplete or poorly executed implementation; lacks coherence or technical effort.
2. Experimental Data & Performance Analysis	Includes strong performance data (e.g., scalability, speedup, bottlenecks); results are well-analyzed and presented using meaningful inputs/outputs.	Includes appropriate performance testing and analysis; inputs and outputs are reasonable and support conclusions.	Minimal performance testing; limited or unclear analysis; lacks depth in exploring inputs or scalability.	Performance analysis is missing or insufficient; little to no data presented.
3. Theoretical Design & Algorithmic Insight	Shows clear algorithmic insight or thoughtful design grounded in theory; builds well on prior work.	Demonstrates a reasonable understanding of algorithmic design and context.	Basic or superficial understanding of algorithms or theoretical background; weak connection to prior work.	Lacks theoretical understanding; no meaningful algorithmic design or research grounding.

**4. Impact &
Contribution
Difficulty**

Tackles a complex or timely problem; results show meaningful technical or academic contribution.

Addresses a problem of reasonable complexity; contributes useful if incremental insights.

Problem is somewhat basic or overused; contribution is limited in scope or relevance.

Problem is trivial, outdated, or lacks any clear contribution or challenge.